Syed Muhammad Taha Hassan Shah

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Summary

Software engineering student passionate about AI & Machine Learning, with frontend development experience, building impactful, data-driven, and well-designed solutions. I am a Sophomore Software Engineering student at UBIT - University of Karachi with hands-on experience in machine learning, frontend development, and cloud technologies. Proficient in Python, JavaScript, TypeScript, C++, and AWS with practical experience building ML models, interactive web applications, and scalable cloud-native deployments. Demonstrated leadership through AWS Cloud Club and competing in the NeurIPS 2025 x Kaggle Google Code Golf Championship, ranking in the top 12% against PhD-level researchers and industry veterans. I am seeking opportunities to apply technical skills and analytical problem-solving in machine learning engineering, AI development, or software engineering positions.

Education

Crescent Bahria Cadet College — Karachi, Pakistan
Matriculation in Computer Science – Percentage: 88.8% — *2020 – 2022*

Bahria College Karsaz — Karachi, Pakistan
Intermediate in Pre-Engineering – Percentage: 72.5% — *2023 – 2025*

UBIT – University of Karachi — Karachi, Pakistan
Bachelor of Software Engineering — *2025 – 2028*

Experience

Machine Learning Engineer — 2025 – Present
Self Employed — *Remote, Karachi, Pakistan*

- Designed and implemented end-to-end machine learning projects using Python, scikit-learn, Pandas, NumPy, and Jupyter Notebooks.
- Performed data preprocessing, cleaning, feature scaling, and exploratory data analysis (EDA) on real-world datasets.
- Currently learning deep learning (neural networks, TensorFlow/PyTorch) to expand into supervised and advanced ML models.
- Familiar with AWS (core services) and Git/GitHub for version control
- Deployed Machine learning models using Streamlit, creating interactive dashboards to visualize clustering results and model predictions.
- Version-controlled all projects using Git and GitHub, following modular and reusable code practices.

Projects

Cardio Risk AI | *Python, scikit-learn, Streamlit, Pandas, NumPy*

- Built a cardiovascular risk assessment system using KNN (K-Nearest Neighbors) machine learning model to predict heart disease risk based on patient health data.
- Deployed the model as an interactive web application using Streamlit, allowing users to input health parameters and receive real-time risk predictions.

Significo-Clone | *HTML, CSS, JavaScript, AWS S3, AWS CodePipeline*

- Built a pixel-perfect static clone of the Significo healthcare website using pure HTML, CSS, and vanilla JavaScript with smooth scrolling and interactive elements.
- Deployed as a fully automated static site on AWS S3 with CI/CD pipeline using AWS CodePipeline + GitHub webhook for zero-downtime updates.

Derivium | *HTML, CSS, JavaScript, Python, Flask, SymPy, Plotly.js, GSAP, KaTeX*

- Built a full-stack calculus visualization web app enabling users to compute limits, derivatives, and partial derivatives with symbolic mathematics backend.
- Integrated interactive 2D/3D graph plotting using Plotly.js with smooth GSAP animations and LaTeX-quality math rendering via KaTeX.

Game Night | *HLDS, AMX Mod X, Pawn, nVault, Linux, FastDL*

- Built a fully customized COD MW4 modded server for Counter-Strike 1.6 from scratch — including engine-level debugging, model precache limit fixes, pdata manipulation, custom zoom logic, and persistent gold economy system.
- Engineered a stable, secure Linux-hosted infrastructure with 50+ plugin ecosystem, FastDL deployment pipelines, and monthly reset architecture — solving crashes, vault corruption, and plugin conflicts through multiple rebuilds.

Hackathons

NeurIPS 2025 x Kaggle - Google Code Golf Championship *Aug 2025 – October 2025*
Google *Remote Hackathon*

- Participated in the Google Code Golf Championship, a featured NeurIPS 2025 competition hosted on Kaggle, competing against 1,142 teams worldwide with participants including PhD-level researchers, industry veterans, and experts with 40+ years of experience in fields like image processing and optimization.
- Achieved final rank of 138 with a peak rank of 109 as an undergraduate student, competing head-to-head against seasoned professionals and academics.
- Developed and iterated multiple submission versions using Python, experimenting with byte-level minimization, creative hacks, and unconventional logic under competitive pressure.
- Collaborated closely with team members to solve problems beyond conventional programming styles, balancing trade-offs between readability, performance, and code size.
- Open-sourced all submissions and experimental approaches to contribute back to the AI & ML community and help future competitors learn optimization strategies.

- Google used this championship's code bases (including my submissions) to train **Gemini AI**, knowing a newer version of Gemini was likely trained on code I wrote is humbling and inspiring.

Volunteering

AWS Cloud Club | Core Team Member *April 2026 – Present*

AWS Cloud Club – Bahria University Karachi Campus *Karachi, Pakistan*

- Collaborate with club leadership to organize cloud computing workshops, technical sessions, and community events focused on AWS technologies.
- Contribute to building a supportive community where students learn cloud fundamentals through practical, hands-on experience.
- Support end-to-end club operations, cross-team coordination, and seamless execution of all activities.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, HTML/CSS
ML & Data Science: scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, TensorFlow, PyTorch
Frontend: React, Next.js, Tailwind CSS, HTML, CSS, JavaScript, TypeScript, GSAP, KaTeX, Plotly.js, APIs (Flask, REST APIs).
Cloud & DevOps: AWS (S3, Lambda, SQS, SNS, EC2, Redshift, RDS, DynamoDB, CloudWatch, Athena, IAM, EventBridge), Vercel, Git, GitHub
Game Server Engineering: HLDS, AMX Mod X, Pawn (.sma), nVault, FastDL, Linux hosting
Developer Tools: VS Code, PyCharm, Git, GitHub, Streamlit
Side Skills: Social Media Marketing, Graphic Design